ModelIt!

Teacher's Guide

Slime Is Serious Science

How Molecules Determine the Properties of Everything You Touch

Quick Reference

Grade Level	Grade 6
NGSS Standard	MS-PS1-1, MS-PS1-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-PS1-1, MS-PS1-2

Develop models to describe the atomic composition of simple molecules and extended structures. Analyze and interpret data on the properties of substances before and after the substances interact to determine if a chemical reaction has occurred.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model showing how molecular size, bond strength, temperature, and concentration determine the physical properties of a substance.
Disciplinary Core Idea	**PS1.A Structure and Properties of Matter** — Substances are made from different types of atoms, which combine with one another in various ways. Atoms form molecules that range in size from simple to complex.
Crosscutting Concept	**Structure and Function** — Students discover that the STRUCTURE of molecules (their size, arrangement, and bonds) directly determines the FUNCTION (physical properties) of the substance.

CAST Testing Alignment

MULTIPLE CHOICE:

Two liquids look identical but one is water (viscosity: 1 cP) and one is glycerol (viscosity: 1500 cP). What molecular difference best explains why glycerol is much thicker?

CONSTRUCTED RESPONSE:

Explain why slime can act like a solid when you hit it quickly but flows like a liquid when you let it sit. Use the concepts of polymer chains and molecular interaction in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

The study of how molecular structure determines material properties is at the heart of chemistry and materials science. Slime is the perfect entry point because it demonstrates that 'solid' and 'liquid' aren't always clear-cut categories.

Polymers Everywhere

Polymers are long chain molecules made of repeating units (monomers). Natural polymers include DNA, proteins, cellulose (wood), and silk. Synthetic polymers include plastics, nylon, rubber, and silicone. The properties of a polymer depend on chain length, branching, and how chains interact with each other.

Non-Newtonian Fluids

Most fluids (water, oil) have constant viscosity regardless of applied force. Non-Newtonian fluids change viscosity under stress. Slime is 'shear-thickening' — it gets MORE viscous under force. Ketchup is 'shear-thinning' — it gets LESS viscous under force (that's why you shake the bottle). These behaviors are determined by molecular structure.

Chemical vs. Physical Changes

Making slime is a CHEMICAL change — the borax creates chemical cross-links between PVA (glue) polymer chains, forming a new substance with different properties. This is different from a physical change (like melting ice) where the molecular structure stays the same but the arrangement changes.

Real-World Applications

Understanding polymer chemistry enables engineers to design bulletproof vests (Kevlar — aligned polymer chains), memory foam (temperature-responsive polymers), biodegradable plastics (polymers that bacteria can break down), and self-healing materials (polymers that can reconnect broken bonds).

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify molecule size, bond strength, temperature, and concentration as the key components determining material properties.

E **Establish Relationships**
Students determine that larger molecules and stronger bonds increase viscosity (positive), while higher temperature decreases viscosity (negative).

V **Visualize & Model**
Students build a model showing how molecular-level interactions produce observable material properties like viscosity and elasticity.

E **Evaluate Outcomes**
Students run liquid vs. perfect vs. solid slime scenarios to observe how changing molecular variables affects physical behavior.

R **Revise & Extend**
Students add cross-linking density or solvent type to explore more complex material behaviors.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Is slime a solid or a liquid? Trick question — it's actually BOTH."*

Watch For

Pull out a container of slime and demonstrate: it pours (liquid!) but bounces when dropped (solid!).

Slide 2: Learning Objectives

3 min

Say: *"Today we're going to figure out WHY slime breaks the rules of matter."*

Watch For

Have students read objectives. Pre-teach 'polymer,' 'viscosity,' and 'non-Newtonian fluid.'

Slide 3: Big Question

3 min

Say: *"Everything you learned about the three states of matter? Slime doesn't care about those rules."*

Watch For

Quick demo: Slowly pour slime (liquid behavior), then punch it (solid behavior). Capture reactions.

Slide 4: LEVER Framework

2 min

Say: *"We'll build a molecular model to crack the slime mystery."*

Watch For

Preview LEVER steps. Emphasize that the answer is at the MOLECULAR level.

Slide 5: Activity 1: Components

8 min

Say: *"What variables at the molecular level affect how slime behaves?"*

Watch For

Guide sorting. Explain that temperature is external while molecule properties are internal.

Slide 6: Activity 2: Connections

8 min

Say: *"Bigger molecules = thicker substance. But higher temperature = more flow. These work against each other."*

Watch For

Demonstrate with cooked spaghetti (tangles = polymers) — short noodles slide easily, long ones tangle.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's dial in the perfect slime — and then make it too thick and too thin!"*

Watch For

Students predict then run three scenarios. Graph and compare viscosity results.

Slide 8: Discoveries

5 min

Say: *"The secret is polymer chains! They tangle when hit, but slide when relaxed."*

Watch For

Connect to real-world non-Newtonian fluids: ketchup, quicksand, body armor.

Slide 9: STEM Challenge

12 min

Say: *"You're now materials scientists. Engineer three different slimes with specific properties!"*

Watch For

Distribute materials. Groups design formulas, test, and measure stretch/bounce/flow.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Temperature
- **INTERNAL:** Molecule Size, Bond Strength, Concentration

Relationships

- **Molecule Size to Viscosity:** POSITIVE (+) — Larger, longer polymer chains get tangled together more easily, making the substance thicker and more resistant to flow.
- **Bond Strength to Viscosity:** POSITIVE (+) — Stronger bonds between molecules hold them together more tightly, resisting flow and making the substance behave more like a solid.
- **Temperature to Viscosity:** NEGATIVE (-) — Higher temperature gives molecules more kinetic energy, causing them to vibrate and move faster. This loosens bonds and reduces tangling, making the substance flow more easily.

Key Discoveries

- Longer polymer chains get tangled together, making the substance thicker and more viscous
- Stronger bonds between molecules mean the substance resists flowing — it acts more solid
- Temperature increases molecular movement, loosening bonds and making substances flow more easily
- Slime is non-Newtonian because its polymer chains tangle when squeezed but slide apart when relaxed

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Matter is always clearly solid, liquid, or gas"	Many real materials don't fit neatly into the three states. Non-Newtonian fluids, gels, plasmas, and amorphous solids (glass) blur the boundaries. The states of matter are useful categories but nature is more complex.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Molecules don't move in solids"	Molecules are ALWAYS moving — in all states of matter. In solids, they vibrate in place. In liquids, they slide past each other. In gases, they fly freely. Temperature determines how much they move. Even at absolute zero, there is still some quantum vibration.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Chemical changes can be easily reversed"	Chemical changes create new substances with new properties. Making slime creates chemical cross-links that can't be undone by stirring or cooling. This is different from physical changes (like ice melting) which are easily reversed. The key test: is a new substance formed?